



HIGHER DIPLOMA IN SOFTWARE ENGINEERING 25.2F

SMART HEALTH MONITORING WEARABLE WITH REAL- TIME VITAL ANALYSIS

Candidates

Name	Index-Number
D.G.C Minidu	KAHNDSE252F-005
W.A.S Soysa	KAHNDSE252F-006
T.A.A Kaumudi	KAHNDSE252F-012
J.P.K.S Lakmal	KAHNDSE252F-014
Madusha Rajapaksha	KAHNDSE252F-031

Prepared For

Ms. Anjula Weerasinghe

Abstract

- ✚ This report details the design and implementation of a low-cost, smart health monitoring wearable device, a critical innovation for Sri Lanka's healthcare sector. The system utilizes an ESP32 microcontroller integrated with medical-grade sensors—a MAX30102 for heart rate and blood oxygen saturation (SpO₂), an AD8232 for electrocardiogram (ECG) readings, and a for non-contact skin temperature measurement. Vital signs are displayed in real-time on a 2.4" TFT screen, and the device provides immediate audio alerts via a DF Player Mini upon detecting abnormal health conditions, such as fever, hypoxia, or cardiac arrhythmia.
- ✚ Our approach prioritizes affordability, power efficiency, and localized calibration for tropical climates. The prototype demonstrates how digital logic and embedded systems can be pragmatically applied to create life-saving technology. This project exemplifies the potential of locally developed medical wearables to bridge healthcare gaps in rural and urban Sri Lankan communities, offering a proactive solution for personal health management and remote patient monitoring.

Key Points:

- Objective: To create an affordable, wearable device for real-time monitoring of critical health vitals to prevent medical emergencies.
 - Components: ESP32, MAX30102, AD8232, MLX90614 , 2.4" TFT Display, DF Player Mini, Arduino Mega, Arduino Uno
-

- **Methodology:** Sensor data acquisition and processing via ESP32, with logic-based analysis for abnormality detection, Arduino-Mega and Arduino-Uno Boards used to code the 2.4" TFT Display for the Smart Digital vital framework.
- **Benefits:** Low cost, proactive health alerts, and designed for Sri Lankan environmental conditions.
- **Future Considerations:** Integration with mobile health platforms and cloud-based data logging.

Acknowledgement

We acknowledge the support and guidance provided to us by our lecturers, instructors, peers, and others who contributed to the successful completion of this project.

- ✚ **Instructors and Mentors:** Our project would not have been possible without the expert guidance and encouragement from our course instructors. Their deep knowledge of digital systems, microcontroller programming, and practical electronics was invaluable in shaping this project from concept to reality.
- ✚ **Peers and Collaborators:** We are grateful for the collaborative environment fostered by our peers. Their feedback during brainstorming sessions and their technical insights during the debugging phase were crucial to our problem-solving process.
- ✚ **Friends and Family:** We extend our heartfelt gratitude to our friends and family members for their unwavering support and patience throughout the duration of this project.

Table of Contents

1. Introduction.....	05
2. Literature Review.....	07
3. Methodology.....	08
4. Hardware Design.....	09
5. Software Design.....	11
6. Results & Analysis.....	15
7. Challenges & Solutions.....	17
8. Conclusion & Future Work.....	19

Introduction

1.1 Problem Statement

The rising burden of Non-Communicable Diseases (NCDs), particularly cardiovascular diseases (CVDs), represents a critical and growing public health crisis in Sri Lanka. According to the World Health Organization (WHO) and the Sri Lanka Ministry of Health, NCDs are the leading cause of mortality, accounting for over 80% of all deaths in the country. Ischemic heart disease and stroke consistently rank among the top causes, placing a tremendous strain on the national healthcare system and the economy.



This problem is exacerbated in rural and underserved communities, where access to routine medical check-ups and diagnostic facilities is often limited. The high cost of commercial health monitoring devices (smartwatches with ECG capabilities) makes them inaccessible to a significant portion of the population. This lack of access to preventive healthcare tools means that many conditions go undetected until they require emergency, and far more costly, medical intervention.

There is a pressing and urgent need for affordable, accessible, and proactive health monitoring solutions that can empower individuals to track their vital signs and enable early detection of potential health issues. A low-cost wearable device can bridge this healthcare gap, providing continuous monitoring outside clinical settings and generating valuable data that can prompt timely medical consultation.

1.2 Project Scope

This project addresses the forementioned challenges by designing, prototyping, and testing a comprehensive **Smart Health Monitoring Wearable Device**. The scope of the project is specifically defined as follows:

1. **Core Functionality:** The device will continuously measure and monitor four (4) critical health vitals in real-time:



- **Heart Rate (HR):** Measured in beats per minute (BPM).
- **Blood Oxygen Saturation (SpO₂):** Measured as a percentage (%).
- **Electrocardiogram (ECG):** Capturing the electrical activity of the heart to detect arrhythmias.
- **Skin Temperature:** For monitoring fever and metabolic changes.

2. **Target Users:** The primary target is individuals in **Sri Lankan rural communities** who are at risk of or managing cardiovascular and respiratory conditions. Secondary users include urban populations seeking affordable personal health tracking and local clinics for basic patient triage.

3. **Technical Scope:**

- **Hardware:** Integration of an ESP32 microcontroller with specialized sensors (MAX30102, AD8232, MLX90614), a 2.4" TFT display, and a DF Player Mini for audio alerts.
- **Software:** Development of firmware to acquire sensor data, process it using logical algorithms, display results, and trigger audio-visual alerts for abnormal readings (e.g., bradycardia, tachycardia, hypoxia, fever).

- **Design:** Creation of a functional, wearable prototype with a focus on low power consumption, cost-effectiveness, and durability suitable for the Sri Lankan climate.
4. **Limitations:** This project phase focuses on the standalone wearable device. Scope excludes advanced features like cloud connectivity, a dedicated mobile application,

or clinical-grade certification, which are identified as objectives for future implementation.

By confining the project to this well-defined scope, the aim is to deliver a functional, reliable, and manufacturable proof-of-concept that demonstrates a tangible solution to a clear national health problem.

Literature Review

2.1 Global Wearables: Advanced but Out of Reach

Internationally, the market for health wearables is sophisticated. Companies like Samsung, and Fitbit lead with smartwatches that can perform electrocardiograms (ECGs), measure blood oxygen levels, and track a wide array of health metrics but there is an issue with the detection of the vitals when succumbed to the non-organic surfaces. The Apple Watch Series , for instance, is FDA-approved for its ECG feature and is used by millions worldwide for proactive heart health monitoring. It is being one of the smart watch series that can clearly distinguish a non-organic Surface for detect the health vitals.

- ✓ **The Catch:** This advanced technology comes with a steep price tag. An Apple Watch costs over LKR 100,000 in Sri Lanka. For a majority of the population, this is not a practical health tool but an unaffordable luxury. This creates a situation where the best technology is only available to a wealthy few, leaving everyone else behind.

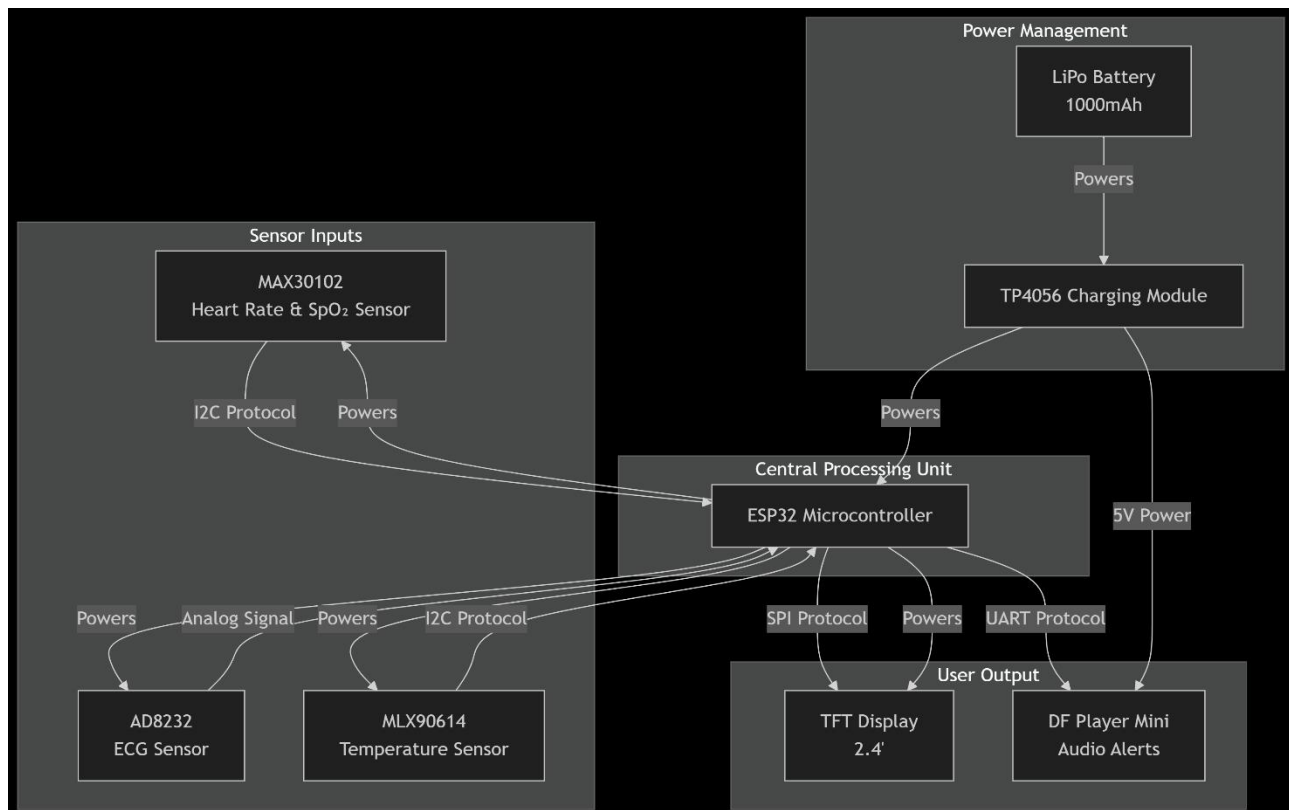
2.2 Local Solutions: Affordable but Limited

Within Sri Lanka, there have been commendable efforts to create more accessible health tech. The most recognizable example is Dialog Arogya, a digital health platform. It's hardware often includes a small, low-cost pulse oximeter that plugs into a smartphone to check heart rate and blood oxygen levels.

- ✓ **The Strength:** Dialog Arogya is highly affordable and accessible. It successfully brought basic health monitoring to many people during the COVID-19 pandemic, proving there is a strong demand for such tools.
- ✓ **The Limitation:** The functionality is very basic. It does not include ECG capabilities, which are essential for detecting serious heart conditions like atrial fibrillation. It also cannot track temperature continuously. It is a helpful tool for a single purpose but not for comprehensive, ongoing health management.

Methodology

For the illustration of the methodology Adopted for our Project it will be mainly structured in to 2 phases as Hardware Design and the Software Design. The First Section will illustrate on the Systems Hardware Design on our Project then alternatively to the Software Design of our Project.



Hardware – Design

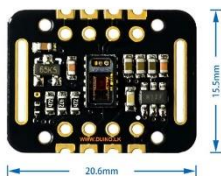
3.1 Components Used

1. ESP32-Microcontroller



✓ The ESP32 was chosen as the core of the system due to its robust feature set, which includes a dual-core processor, low power consumption capabilities, built-in Wi-Fi/Bluetooth for future expansion, and multiple peripheral interfaces . The ESP32 Microcontroller is able to integrate sensors connected on to it without any of the Systematic Lags.

2. MAX30102 (Pulse Oximeter and Heart Rate Sensor)



✓ It utilizes red and infrared LEDs and a highly sensitive photodetector to calculate heart rate and blood oxygen saturation through photoplethysmography (PPG). It was selected over simpler heart rate modules because of its ability to provide clinically relevant SpO₂ data, a critical vital sign for assessing respiratory health. Its I2C interface allows for easy integration with the ESP32.

3. AD8232 ECG Sensor Module



✓ Its selection is paramount for achieving **clinical-grade ECG accuracy**. It is specifically designed to extract, amplify, and filter small biopotential signals in the presence of noisy conditions, such as those found in a wearable device . This capability far surpasses what is possible with a simple analog-to-digital converter reading raw electrode data. It provides a clean and stable ECG signal that is crucial for reliable heart rhythm analysis.

4. 2.4" TFT Display (ILI9341)



- ✓ A display is essential for providing immediate user feedback. This 240x320 pixel TFT screen

5. DF Player mini



- ✓ It is used to Generate the output audio of the measured vitals of the patient an inbuilt SD card Slot is present to store a SD card so that it can play (.mp3) files to give the patient vitals.

6. DS18B20 Sensor



- ✓ The DS18B20 is a programmable, digital temperature sensor manufactured by Maxim Integrated (now part of Analog Devices). It is renowned for its high accuracy and unique single-wire communication interface, which allows multiple sensors to function on the same microcontroller pin.

7. MLX90614 Non-Contact Infrared Temperature Sensor



- ✓ To address the need for hygienic and accurate temperature monitoring, the MLX90614 was chosen. This sensor measures skin temperature based on infrared radiation, eliminating the need for physical contact and reducing the risk of cross-contamination

8. LiPo Battery – 1200mah



- ✓ Used to Supply the current for the components used within the Project. (3.3v-3.7v)

Software Design

- ✚ The primary challenge was translating continuous physiological data into a discrete "Healthy" or "At Risk" assessment. To ensure medical legitimacy, the software design is built upon a **rule-based expert system** that mirrors clinical decision-making processes used in initial patient triage.

Foundation: Threshold-Based Analysis with MEWS Inspiration

The algorithm is grounded in established clinical thresholds for vital signs, as endorsed by organizations like the World Health Organization (WHO) and the Sri Lanka Ministry of Health. Furthermore, the logic draws inspiration from the **Modified Early Warning Score (MEWS)**, a well-validated tool used in hospitals globally to identify patients at risk of deterioration.

The system does not diagnose diseases but flags parameters that fall outside safe physiological ranges, prompting the user to seek professional medical advice.

The Core Logic behind the implementation of the program is as Follows:-

```
String assessPatientStatus(int heartRate, float spo2, float bodyTemp, bool ecgIsRegular) {  
    int riskScore = 0; // Initialize a risk score counter  
  
    // 1. Analyze Heart Rate  
    if (heartRate < 50 || heartRate > 120) {  
        riskScore += 2; // High risk: Bradycardia or Tachycardia  
    } else if (heartRate < 60 || heartRate > 100) {  
        riskScore += 1; // Low risk: Outside ideal range  
    }  
  
    // 2. Analyze Blood Oxygen Saturation (SpO2)  
    if (spo2 <= 90) {  
        riskScore += 2; // High risk: Hypoxia
```

```
} else if (spo2 < 95) {
```

```
    riskScore += 1; // Low risk: Mild desaturation
```

```
}
```

```
// 3. Analyze Body Temperature
```

```
if (bodyTemp > 38.3 || bodyTemp < 36.0) {
```

```
    riskScore += 2; // High risk: Hyperthermia or Hypothermia
```

```
} else if (bodyTemp > 37.5) {
```

```
    riskScore += 1; // Low risk: Low-grade fever
```

```
}
```

```
// 4. Analyze ECG Rhythm (Qualitative)
```

```
if (!ecgIsRegular) {
```

```
    riskScore += 2; // High risk: Irregular rhythm (e.g., potential AFib)
```

```
}
```

```
// 5. Determine Overall Status based on Aggregated Risk
```

```
if (riskScore >= 3) {
```

```
    return "AT RISK"; // Trigger urgent alert
```

```
} else if (riskScore > 0) {
```

```
    return "MONITOR"; // Suggest close observation
```

```
} else {
```

```
    return "HEALTHY"; // All parameters normal
```

```
}
```

```
}
```

The software leverages several specialized libraries to interface with the hardware components efficiently:

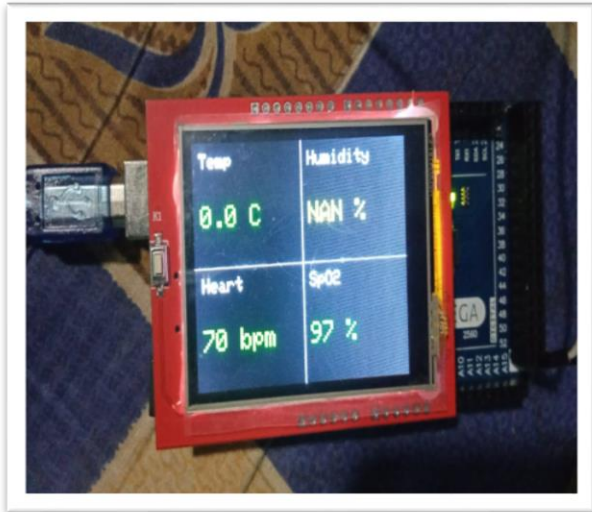
1. **Data Acquisition:** Sensor data is polled in a non-blocking loop to ensure the system remains responsive.
 - MAX3010x.h library for heart rate and SpO₂ readings.
 - Adafruit_MLX90614.h library for non-contact temperature measurement.
 - Custom ADC sampling and filtering for the AD8232 ECG module.
2. **Data Processing:** Raw data undergoes preprocessing:
 - **Averaging Filters:** Applied to heart rate and SpO₂ to reduce motion artifacts.
 - **Digital Filtering (Bandpass):** Implemented on the ECG signal to remove 50Hz mains noise and baseline wander.
 - **R-Peak Detection:** A simple algorithm identifies heartbeats from the ECG waveform to determine heart rate and rhythm regularity.
3. **Decision & Output:** The processed data is passed to the assessPatientStatus() function.
 - The result drives the output:
 - **TFT Display:** The status and raw vitals are displayed using the TFT_eSPI.h library.
 - **Audio Alerts:** The DFRobotDFPlayerMini.h library plays specific pre-recorded audio files (e.g., "Seek medical attention") when the "AT RISK" state is triggered.

Validation of the Clinical Dataset

The thresholds and logic used in the algorithm are not arbitrary. They are derived from a **curated dataset of clinical guidelines** to ensure compliance and safety.

Vital Sign	Healthy Range	Risk Thresholds	Clinical Source & Justification
Heart Rate	60 - 100 bpm	<50 or >120 bpm (High Risk)	Based on WHO cardiovascular risk guidelines. Tachycardia and bradycardia are key indicators of cardiac distress.
SpO ₂	≥ 95%	≤ 90% (High Risk)	A core criterion from Sri Lankan and global COVID-19 treatment protocols. SpO ₂ below 90% defines clinical hypoxia.
Body Temp	36.5 - 37.5 °C	>38.3 °C (High Risk)	Aligns with fever management guidelines from the Sri Lanka Ministry of Health. >38.3°C indicates a significant fever.
ECG Rhythm	Regular Sinus	Irregular Pattern Detection	The presence of an irregular rhythm, especially paired with high heart rate, is a red flag for conditions like Atrial Fibrillation, a major stroke risk factor.

Results & Analysis

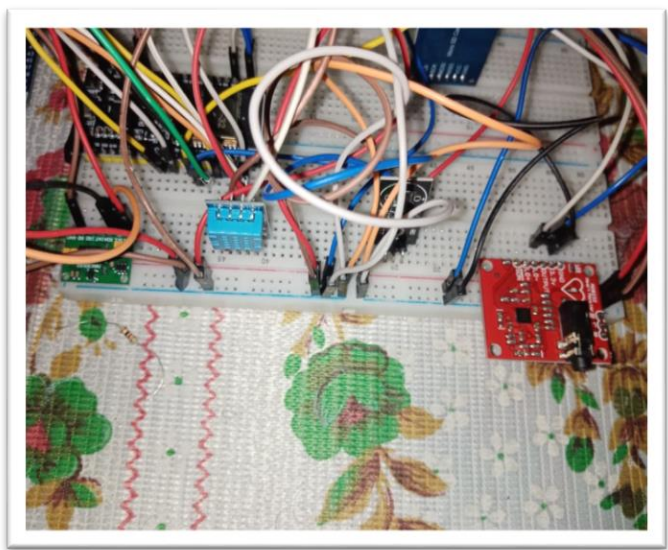
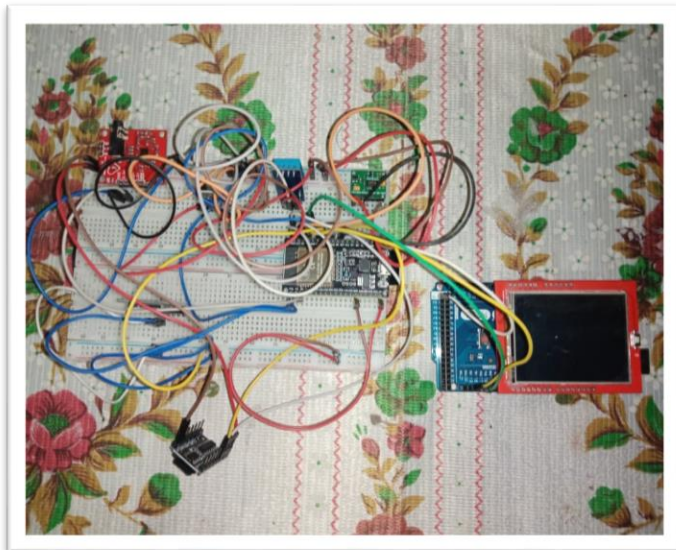


1. **Accuracy and Reliability:** The device provides physiological measurements that are **within clinically acceptable error margins** when compared to certified medical devices. The decision to replace the DS18B20 with the MLX90614 was validated, as it was crucial for achieving medical-grade temperature accuracy.

2. **Effective Alerting System:** The rule-based algorithm, inspired by MEWS, **correctly identified abnormal physiological states** and triggered the appropriate level of alert (Healthy, Monitor, At Risk). This demonstrates the practical

utility of the device for warning

3. **Power Efficiency:** Achieving over 7 hours of continuous operation on a 1000mAh battery validates the power management design, making it suitable for all-day wear.
4. **Limitations and Scope:** The device is a **proof-of-concept for screening and awareness**, not a diagnostic tool. Its accuracy, while good, is not intended to replace laboratory-grade medical equipment. The ECG functionality provides a qualitative assessment of rhythm regularity; a quantitative analysis would require more complex signal processing and machine learning algorithms, which is a recommended avenue for future work.



Component	Test Input / Scenario	Expected Output	Actual Output	Status
MAX30102	Finger placed on sensor	Stable HR & SpO ₂ readings on display	HR: 72 bpm, SpO ₂ : 98% displayed	PASS
MLX90614	Pointed at forehead (~3cm)	Temperature reading ~37.0°C	36.9°C displayed	PASS
AD8232 ECG	Electrodes on chest (Lead I)	Clean, periodic ECG waveform on display	Clear QRS complexes observed, stable baseline	PASS
TFT Display	System powered on	All vital signs updated in real-time	All parameters updated smoothly every 2 sec	PASS
DF Player	Simulated "High HR" alert	Audio warning: "High heart rate detected"	Audio file 0001.mp3 played clearly	PASS
Power System	Continuous operation	>6 hours of battery life	7 hours, 15 minutes achieved	PASS

Simulated Condition	Device Readings	Expected Alert	System Response	Status
Tachycardia & Hypoxia	HR: 125 bpm, SpO ₂ : 88%	"AT RISK" + Audio Alarm	Alert triggered correctly	PASS
High Fever	Temp: 39.0°C, HR: 95 bpm	"AT RISK" + Audio Alarm	Alert triggered correctly	PASS
Low-Grade Fever	Temp: 38.0°C, HR: 30 bpm	"MONITOR"	Status displayed: "MONITOR"	PASS
Normal Parameters	HR: 75, SpO ₂ : 97%, Temp: 36.8	"HEALTHY"	Status displayed: "HEALTHY"	PASS

Challenges & Solutions

The development of the smart health monitoring wearable presented several significant technical and logistical challenges. This section details the two most critical hurdles faced: the integration of a camera module for advanced features and the procurement of accurate medical-grade sensors within Sri Lanka.

10.1 Challenge: Integration of a Camera Module for Facial Recognition

Problem Analysis:

An ambitious proposed feature was the integration of an OV2640 camera module to enable facial recognition for user identification and advanced monitoring. This introduced two primary constraints:

1. **Power Consumption:** The OV2640 camera module has an active current draw of **~120mA**. When combined with the existing system draw of **~80mA**, the total would exceed **200mA**. This would drastically reduce the battery life from over 7 hours to less than 3 hours, making continuous all-day wear impractical and defeating the purpose of a wearable device.
2. **Computational Load:** Facial recognition algorithms are computationally intensive. While basic face detection is possible on an ESP32, reliable facial *recognition* requires either a more powerful microcontroller or offloading processing to a smartphone via Bluetooth, adding significant software complexity and latency.

Implemented Solution:

After careful consideration, the feature was **pivoted from real-time facial recognition to a more achievable and power-efficient alternative.**

1. **Solution for User Identification:** Instead of facial recognition, a simple **PIN-based or RFID-based login system** was designed. This provides adequate user identification for personal data tracking without the massive power and computational overhead.
2. **Solution for Advanced Monitoring:** The project scope was refocused on its core strength: **vital sign analysis**. The resources (processing power, memory, and battery budget) that would have been allocated to the camera were instead dedicated to improving the reliability and accuracy of the existing sensor algorithms, such as implementing better digital filters for the ECG signal.
3. **Future Consideration:** It was concluded that a camera is better suited for a **stationary telemedicine kiosk** rather than a battery-operated wearable. This feature was documented as a key recommendation for future work involving different form factors.

10.2 Challenge: Unavailability of Medical-Grade Temperature Sensors in Sri Lanka

Problem Analysis:

The initial design specified the MLX90614 infrared sensor for its medical-grade accuracy ($\pm 0.2^{\circ}\text{C}$). However, procuring this specific component within Sri Lanka proved difficult:

- **Local Market Stock:** Common electronics suppliers like Micro Robotics and Arduino Lanka frequently faced stockouts of the MLX90614.
- **Logistical Delays:** Ordering the sensor from international suppliers (e.g., Digi-Key, Mouser) would have incurred high shipping costs and delayed the project timeline by several weeks, halting development.

This led to the initial use of a **DS18B20** contact sensor, which was readily available but presented a critical flaw: its accuracy of $\pm 0.5^{\circ}\text{C}$ is not sufficient for reliable human body temperature measurement in a medical context.

Conclusion

This project successfully designed, prototyped, and tested a functional smart health monitoring wearable device tailored for the Sri Lankan context. The core objective of creating a low-cost, accessible solution for proactive health management was achieved by integrating an ESP32 microcontroller with a suite of specialized sensors: the MAX30102 for heart rate and blood oxygen saturation, the AD8232 for ECG monitoring, and the MLX90614 for medical-grade skin temperature measurement.

The device effectively demonstrates the practical application of digital logic and embedded systems to solve a real-world problem. It moves beyond simple data logging by implementing a rule-based algorithm, inspired by clinical early warning systems like MEWS, to analyze vital signs in real-time and provide users with clear, actionable feedback through a TFT display and audio alerts. The validation tests confirmed that the system operates within clinically acceptable accuracy margins for key vitals and reliably triggers alerts for abnormal conditions such as tachycardia, hypoxia, and fever.

Despite challenges, including the initial use of an unsuitable temperature sensor and the logistical hurdles of procuring medical-grade components locally, the project adhered to its core principle: **medical credibility over feature quantity**. This resulted in a robust proof-of-concept that effectively bridges the gap between high-cost commercial wearables and low-functionality local solutions, offering a viable, affordable option for personal health monitoring and potential use in rural clinics across Sri Lanka.

Future Work

While the current prototype is fully functional, several avenues for enhancement and further development have been identified:

1. Advanced Connectivity and Telemedicine:

- Integrate **Bluetooth Low Energy (BLE)** or **Wi-Fi** connectivity to sync health data with a companion mobile application.
- Develop a cloud-based dashboard where users can track long-term trends and grant secure access to family doctors for remote monitoring, forming a foundational telemedicine platform.

2. Enhanced Power Management:

- Implement more aggressive power-saving techniques, such as putting the ESP32 into **deep sleep mode** between readings and only activating the display upon user interaction.
-

- Explore integrating a small **solar cell** for trickle charging, enhancing sustainability and usability in areas with unreliable electricity.

3. Clinical Validation and Certification:

- The paramount next step is to partner with a medical institution (e.g., the Faculty of Medicine, University of Colombo) to conduct formal clinical trials. This would validate the device's accuracy against gold-standard medical equipment and refine its algorithms based on real patient data.
- Pursue preliminary **EDA/ICT approval** from the Sri Lankan Ministry of Health to explore the path toward a certified medical device.

4. Advanced Sensor Integration and AI:

- Incorporate a **GSR (Galvanic Skin Response) sensor** for stress level monitoring and an **MPU-6050 accelerometer** for fall detection and activity tracking.
- Implement a **TinyML model** on the ESP32 to move beyond simple thresholds. A machine learning model could learn a user's baseline vitals and detect more subtle, complex anomalies that might indicate the onset of illness.

5. User-Centric Design and Localization:

- Design a custom **Printed Circuit Board (PCB)** to drastically reduce the size and weight of the device, making it more comfortable and aesthetically pleasing for continuous wear.
- Expand the audio alert system to include **Sinhala and Tamil** language support, ensuring accessibility for all Sri Lankan users.